

Course Code: CSE3003	Course Title: Computer Networks	TPC	3	2	4
Version No.	2.0				
Course Pre-requisites/ Co-requisites	CSE2005				
Anti-requisites (if any).	SWE3001, ECE3003				
Objectives:	1. To introduce basic concepts of Computer Networks 2. To expose state-of-the-art technologies in computer network protocols, architectures, and applications.				
CO's Mapping with PO's and PEO's					
Course Outcomes	Course Outcome Statement	PO's / PEO's			
CO1	Independently understand basic computer network technology and Data Communications System and its components.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8			
CO2	Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.	PO1, PO2, PO3, PO4, PO5, PO6, PO7			
CO3	Analysing packet formats of TCP, UDP, IP, MAC etc.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8			
CO4	Identify the different types of network devices and their functions within a network	PO1, PO2, PO3, PO4, PO5			
CO5	Understand and build the skills of subnetting and routing mechanisms.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8			
CO6	Familiarize with the basic protocols of computer networks, and how they can be used and implement	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8			
TOTAL HOURS OF INSTRUCTIONS: 45					
Module No. 1	Introduction				6 Hours
Computer network and its history, progress and application, Internet, Network architecture, Networking devices. OSI Model, TCP/IP Protocol stack, Networking in different OS.					
Module No. 2	Physical Layer				8Hours
Guided media, Unguided media, Network Topologies, Analog and digital communication. Encoding mechanisms, Packet Switching, Circuit Switching.					
Module No. 3	Data Link Layer				8 Hours
Framing, HDLC, PPP, Error detection, Error Correction, MAC Protocols, Reliable Transmission, Ethernet, 802.3, 802.5, 802.11.					
Module No. 4	Network Layer				7 Hours
IP addressing schemes, IPV4, Subnetting, IPV6, shift from IPV4 to IPV6, ICMP, DHCP, ARP. Routing Protocols: Distance-vector and link-state routing. RIP, OSPF, BGP.					
Module No. 5	Transport Layer				8 Hours
Connection Oriented and connection less service, TCP and UDP, Port Addressing, Remote Procedure Call, Flow Control vs Congestion Control, Quality of Service.					
Module No. 6	Application Layer Protocols				8 Hours
Application Layer Protocols: World wide web and HTTP, HTTPS, Domain names: DNS, File Transfer: FTP, Electronic mail: SMTP, Peer to peer networking, Torrent, VPN. Session management, Data compression techniques.					

Text Books

1. Behrouz A. Forouzan , “Data Communications and Networking”, Fourth Edition TMH, 2017.

References

2. Andrew S Tanenbaum, “Computer Networks”, 6th Edition. Pearson Education/PHI, 2021.
3. Ying-Dar Lin, Ren-Hung Hwang, Fred Baker, “Computer Networks: An Open Source Approach”, Mc Graw Hill Publisher, 2012s.
4. Larry L. Peterson, Bruce S. Davie, “Computer Networks: A Systems Approach”, Morgan Kaufmann Publishers, 6th Edition, 2021.
5. William Stallings, “Data and Computer Communication”-, 10th Edition, Pearson Education, 2017.

Lab Exercises**Use Cisco Packet Tracer/ NetSim/Wireshark**

1. Design of Topology using network devices.
2. Design and Configuring LAN, WAN and VLAN
3. Analysis of DNS Server and DHCP.
4. Router Serial Point to Point Connection HDLC, PPP with PAP and CHAP.
5. Port-Security Configuration to prevent MAC flooding attack
6. Design and Configuring WAN using three Routers using serial DCE cables with Static Routing.
7. Design a network using fixed length and Variable length Subnetting for a class C Ipv4 address and configure it in
8. Router.
9. Simulation & Analysis of Transport Layer Protocol (TCP)
10. Understanding TCP Congestion Control Mechanism
11. Simulation & Analysis of RIP Routing Protocols:
12. Simulation & Analysis of OSPF Routing Protocols
13. Understanding Working and configuration of a Switched Ethernet.
14. Understanding and Simulation of select Network Protocols using Open Source Simulator(s).
15. Analyzing FTP Traffic
16. 15. Analyzing HTTP Traffic

Course Type	Embedded Theory and Lab(ETL)		
Mode of Evaluation	Theory		75%
	Continuous Assessment Test-1	15	
	Continuous Assessment Test-2	15	
	Digital Assignments/Quizes (Min)	30	
	Final Assessment Test	40	
	Laboratory		25%
Modified by	Dr. Prabha Selvaraj and Dr. Sunil Kumar Singh		
Recommended by the Board of Studies on	14th BoS, 11.05.2024		
Date of Approval by the Academic Council	12th Academic Council, 25.05.2024		